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Exploring non-iterative time trade-off methods for valuation of EQ-5D-5L health states

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Abstract

Introduction The composite time trade-off (cTTO) method is used as the primary method for valuing EQ-5D-5L health states, but it requires intensive interviewer training and stringent quality control, which increases the burden of conducting cTTO studies. In this study, two non-iterative variants of the TTO method, non-stopping TTO (nTTO) and open-ended TTO (oTTO), were tested head-to-head with the cTTO method aiming to reduce the administration burden.

Methods 31 EQ-5D-5L health states from an orthogonal array was selected and valued by a general public sample in China. Data were collected by 7 interviewers with all interviewers performed an equal number of interviews using all three TTO methods. We compared the value distribution, logical consistency, administration burden, and modeling performance of these three TTO methods.

Results In total, 422 participants participated in the valuation interviews, with 139 using the nTTO method, 140 using the oTTO method, and 143 using the cTTO method. Both oTTO and nTTO methods saved around 10 min for conducting an interview. The mean values of three methods were similar with each method showed different characteristics in their value distributions. cTTO outperformed the other two methods in terms of modeling performance.

Discussion Both non-iterative TTO methods showed potential for valuing EQ-5D health states, although their data distributions and modeling performance were inferior to the cTTO method. The results of this study showed the potential of these two alternative non-iterative TTO methods.

Keywords EQ-5D-5L · Time trade-off · Value set · Data collection · Quality control

Introduction

The EuroQol Group established value sets for the EQ-5D-5L using an international protocol called the EuroQol Valuation Technology (EQ-VT) Protocol [1]. This protocol utilizes both the composite time trade-off (cTTO) and discrete choice experiment (DCE) methods to gather preferences from a representative sample of the general public [2]. The cTTO method provides cardinal values, but requires in-person interviews, which may affect the quality of data due to interviewer performance [3]. To address this, an intensive quality control (QC) process was implemented [4]. On the other hand, the DCE method is easier to administer but does not produce direct utility values, requiring inferences from participant choices. While some studies have combined data from both methods for estimating EQ-5D value sets [5], others have solely relied on cTTO data [6, 7].

The cTTO method has its own challenges, such as data clustering, where data collected using this method tends to

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cluster around certain values [8, 9]. This clustering occurs in part because some respondents' responses are influenced by the starting value of the cTTO task, leading them to finish the task prematurely [10, 11]. Despite efforts to improve interviewer performance and respondent engagement in the task, no improvements have been made to the TTO method itself.

As the primary elicitation method in the EQ-VT protocol, the cTTO design includes intensive interviewer training and a regular data collection QC process. These requirements make conducting a cTTO valuation study less feasible, but failing to comply with these procedures reduces the validity of the cTTO data. Valuation studies aiming to establish national value sets should strictly follow the EQ-VT protocol for optimal data, but smaller, methodological, and exploratory studies are often less stringent, leading to a divide between values obtained from well-planned valuation studies and values produced in other contexts. Therefore, there is a need to adapt the current cTTO method for easier application, such as online administration, which is not possible with the current cTTO method included in the EQ-VT protocol [12].

To address the challenges of cTTO data and develop a more user-friendly TTO method, this study tested two TTO variants: the non-stopping TTO (nTTO) [13] and the open-ended TTO (oTTO) [14]. The nTTO, which was first tested in Singapore and Norway [13], requires respondents to state preferences for a set of alternatives in the same order without stopping, even if they have stated indifference. This design precludes satisficing behavior and was found to be less time-consuming and prone to data clustering compared to cTTO. The performance of the nTTO, however, differed in Norway and Singapore, suggesting that nTTO may be susceptible to interviewer or cultural effects. Therefore, this study aimed to further test nTTO variant in a different population.

The oTTO has been widely used in clinical research. For example, it was first used to elicit the preference of parents taking care of their children with Autism Spectrum Disorders [14]. The oTTO asks respondents to state their indifference point with minimal prompting and can be administered without an interviewer, such as via postal surveys [14]. Nevertheless, compared to the TTO methods such as cTTO developed in the context of health-state valuation, the commonly used oTTO methods were less rigorous; for example, no computer-assisted software was used, no worse-than-dead scenario was considered, and the time frame was set to 40 years which is subjected to time discounting. Therefore, we tested an oTTO design which is similar to cTTO in terms of time frame and visual aids, to compare their performances. The aim of this study was to evaluate whether these two alternative TTO methods are easier to apply and examine whether they could produce TTO values that are comparable with the cTTO method.

Methods

Health-state selection and sample size

To compare the performance of the two TTO variants with the cTTO method, 31 EQ-5D-5L states were selected and divided into two sets of 16 health states. This design represented a range of health states with varying severity and allowed for the analysis of a main effects model [15]. According to the EQ-VT protocol, a sample size of 100 observations per state is recommended for estimating national value sets [16]. However, as this study was not focused on establishing national value sets, the sample size was reduced to 70 observations per state. The total sample size was 420, consisting of 70 observations per state, two sets of health states, and three different methods (70 observations * 2 sets * 3 methods).

The TTO methods

The cTTO method has been described in detail elsewhere [1]. In this study, we will only provide a brief overview of the two non-iterative TTO methods. Both methods share the same life length (10 years in the better-than-dead scenario and 20 years in the worse-than-dead scenario) and life description (Life A = full health, Life B = impaired health) as the cTTO method, but differ in the manner in which the length of Life A is presented and iterated.

In the nTTO method, Life A always starts with a length of 10 years and respondents are presented with three options: Life A is better, A and B are the same, or Life B is better. The questioning continues, regardless of the response, and the length of Life A decreases by one year after each choice. If the nTTO value is estimated either to be above 0.95 or below 0, additional questions are asked. If the value is 0.95, up to four durations (9 years and 6 months, 9 years and 9 months, 9 years and 10 months, and 9 years and 11 months) are asked until the preference changes from Life B to Life A. If the preference suggests a value of less than 0 (i.e., worse than dead), the task continues by asking about the respondent's preferences for 11 durations, in descending order, in the lead-time framework of EQ-VT (where Life B is 10 years in full health followed by 10 years in an impaired health state). Table 1 outlines the process for calculating TTO values from nTTO responses.

As discussed in the introduction part, the oTTO was first used in clinical setting. To compare its performance with the cTTO method, we decided to align its design and presentation with the current EuroQol cTTO method, for example, by constraining the life length to 10 years instead

Table 1 TTO values for nTTO method

Response pattern	nTTO value is determined by ...	Example
'No preference' is indicated only once	The corresponding life years	'No preference' for 3 years of Life A only → TTO value = 0.3
'No preference' is indicated multiple times	The midpoint of the minimum and maximum life years	'No preference' for 3, 4 and 5 years of Life A → TTO value = 0.4
No preference' is not indicated	The midpoint of the maximum life years for 'prefer Life B' and the minimum life years for 'prefer Life A'	'Prefer Life A' when Life A is 5 to 10 years but 'Prefer Life B' when Life A is 0 to 4 years → TTO value = 0.45
Response pattern is not meaningful	Dropped	'Prefer Life A' when Life A is 5 years but 'Prefer Life B' when Life A is 6 years; 'No preference' for all 11 durations

of 40 years, by adding a worse-than-dead scenario and by designing a software that the visual elements are identical to cTTO method. The core design of the oTTO method was kept, that is, two preliminary TTO questions were asked followed by an open-ended question aimed at determining the equivalent length of life in full health to 10 years in the target state. The oTTO uses a hypothetical scenario in which the respondent is living with impaired health (Life B) and has the option of taking a medicine that would restore their health to full health (Life A) but shorten their lifespan. The consideration of framing the question as 'taking a medicine' instead of other possible framings is based on the assumption that medicine use is less intrusive and has been experienced by most respondents. In the first TTO question, the reduction in life expectancy was a randomly selected number between 2 and 7 years, and the respondent is asked to choose whether to take the medicine or not (the option of "A and B being the same" is not allowed). The design of randomizing the starting point was motivated by previous study of Liv et al., who founded the TTO values are biased by the anchoring effect [10]. The choice of 2–7 years was considered not to extreme, i.e., not toward the end of the scale of 10 years and was considered diverse enough to mitigate the anchoring effect, if any. For the second question, it was considered to give more extreme values toward either end of the scale so that preferences of these values are probed. Therefore, based on the response to the first question, the second TTO question either shorten Life A to 1 year (if the respondent choose to take the medicine) or increase it to 9 years (if the respondent choose not to take the medicine). If the respondent indicates that the health state is worse than dead, the task continues with the EQ-VT lead-time TTO. The TTO value in the oTTO is calculated from the answer to an open-ended question to identify the point at which the reduction in life expectancy would be excessive, such that they would prefer to remain in Life B for 10 years. Appendices 1 and 2 provided the screenshots of nTTO and oTTO methods, respectively.

The data collection for the cTTO method was conducted using EQ-PVT, a PowerPoint replica of the EQ-VT software, and a newly developed online software was created to collect data for the oTTO and nTTO methods. The online software was designed to mirror the format and color scheme used in the EQ-PVT software. The interviewer script for the oTTO and nTTO methods was adapted from the cTTO interviewer script used in EQ-VT studies. The software and interviewer script were pre-tested with a sample of 12 participants (6 participants for each method), followed by a pilot study with a sample of 40 participants (20 participants for each method).

Interview and data collection procedure

We recruited 7 undergraduate students from Guizhou Medical University to serve as interviewers and provided them with a comprehensive one-day training on conducting the cTTO interviews. The total sample size for the study was 420 participants, with each interviewer being assigned to conduct 60 interviews. To reduce the impact of interviewer effects, each interviewer conducted 20 interviews using each of the TTO methods in a randomized order to control for any possible ordering effects. Before beginning formal data collection, each interviewer was required to conduct 3 practice interviews for each method.

To ensure a representative sample of the general population, four quotas were established based on gender, age, education attainment, and urbanicity (Hukou). Participants were recruited from Guizhou and Chongqing and were interviewed face-to-face to value one block of health states using a single TTO method. Before the TTO tasks, demographic information and self-reported EQ-5D-5L data were collected. The cTTO tasks were performed following the EQ-VT protocol V2.0 [2] and with a QC process [4]. QC reports were also produced for the nTTO and oTTO methods. However, given the minimal time needed for explaining the TTO task using wheelchair example and the time needed to value a single EQ-5D-5L state are not known for nTTO

and oTTO methods, we only provided feedback on another two important QC criteria, i.e., whether the worse-than-dead is explained and whether there are any inconsistency [4]. In addition to these two criteria, any sign of interviewer effects were also reported back to the interviewers.

Data analysis

We compared the performance of two non-iterative TTO methods, nTTO and oTTO, to the benchmark cTTO method, based on value distribution, logical consistency, administration burden, and modeling performance. To facilitate a better comparison, we rounded the nTTO values to 0.05, as the nTTO method was designed to elicit more detailed values at the upper end of the scale.

1). *Administration burden*—The burden of administering the TTO methods was assessed by the amount of time spent on the valuation tasks. Specifically, we measured the time spent on the three practice states, 15 or 16 official states, and the total time spent on both. A shorter time was considered to be more efficient.

2). *Logical inconsistency*—The quality of the TTO values was assessed at the individual level using the frequency of logical inconsistencies. This was defined as the number of instances where a participant gave a higher value to a worse health state [3]. Logical inconsistencies were counted as instances where the difference in values was greater than 0, with severe logical inconsistencies defined as differences greater than 0.5. This measurement was used as a key criterion in the QC process [4] and was used to evaluate the data quality of the study.

3). *Value distribution*—We analyzed the TTO data both at an aggregate and individual level for each method. At the aggregate level, we presented the average values for all health states, the percentage of negative values, and visualized the data using histograms. We also identified and counted any “spikes” (defined as values selected by more than 5% of participants) and “gaps” (defined as values selected by less than 1% of participants) in the histograms. At the individual level, we determined the number of unique TTO values for each participant and the range of values they provided.

4). *Modeling performance*—We applied a cross-attribute level effects (CALE) model [17, 18] to analyze the data obtained from each TTO method. By capitalizing on the similarity in level labels across EQ-5D-5L dimensions, CALE models reduce the number of model parameters. In terms of EQ-5D-5L, an 8-parameter model can be constructed including 5 dimension parameters, each representing the level-5 problems of a different dimension (MO for mobility, SC for self-care, UA for usual activities, PD for pain/discomfort, AD for anxiety/depression), and 3 level parameters for ‘slight’ (L2), ‘moderate’ (L3), and ‘severe’

(L4). The model is based on the assumption that the ratios of disutility associated with level-2 to level-4 problems versus level-5 problems are constant across dimensions. Thus, to derive the disutility of an intermediate level, one needed to multiply the corresponding level parameter (L2, L3, or L4) with the dimension parameter (MO, SC, UA, PD or AD). For example, the disutility of having moderate problem in usual activities is calculated by multiplying UA with L3. This model specification has been widely tested in various EQ-5D-5L valuation datasets and proven to be an effective model for EQ-5D-5L TTO data analysis [7, 19]. We compared the (1) face validity of the models, i.e., the significance and monotonicity of the coefficients; (2) model fit, i.e., Akaike information criterion (AIC), Bayesian information criterion (BIC) and adjusted R-squared values of the models; (3) prediction performance, i.e., by TTO method, we calculated the mean squared errors (MSE) of the models in predicting the observed values of all health states, with a higher MSE indicating large prediction errors [20].

Results

The data collection was carried out between September 2019 and September 2020 in Guizhou and Chongqing, China. Seven interviewers each conducted approximately 20 interviews per method, totaling around 60 interviews per interviewer. The order in which the methods were used varied among the interviewers. Three interviewers started with the oTTO method, followed by nTTO and cTTO, while the other three used the nTTO method first, followed by oTTO and cTTO. One interviewer used the cTTO method first, before nTTO and oTTO. In total, 422 participants participated in the valuation interviews, with 139 using the nTTO method, 140 using the oTTO method, and 143 using the cTTO method. The sample characteristics were similar across the three methods, except that the participants who completed the cTTO tasks were more likely to have a rural Hukou status (Table 2).

Table 3 summarizes the administration burden, logical consistency, and value distribution of the three TTO methods. On average, the time spent on both practice tasks and the official tasks was less for the two non-iterative TTO methods (nTTO and oTTO) compared to the cTTO method. The average time taken to complete the practice tasks was approximately 2–3 min shorter when using the nTTO and oTTO methods, with the shortest time spent being on the oTTO method. Similarly, both non-iterative methods required less time for the official tasks, resulting in a time savings of around 10 min for the entire interview (practice and official tasks combined). In terms of logical consistency, both non-iterative TTO methods showed a higher frequency

Table 2 Demographics of the sample across the three TTO variants

Demographic variables	nTTO <i>n</i> = 139 <i>n</i> , %	oTTO <i>n</i> = 140 <i>n</i> , %	cTTO <i>n</i> = 143 <i>n</i> , %
<i>Gender</i>			
Male	71, 51.08	69, 49.28	71, 49.65
Female	68, 48.92	71, 50.72	72, 50.35
<i>Age group</i>			
18–29	28, 20.14	29, 20.71	31, 21.68
30–39	27, 19.42	29, 20.71	28, 19.58
40–49	28, 20.14	26, 18.57	28, 19.58
50–59	28, 20.14	29, 20.71	28, 19.58
> 60	28, 20.14	27, 19.29	28, 19.58
<i>Hukou*</i>			
Urban	79, 56.83	74, 52.86	70, 48.95
Rural	60, 43.17	66, 47.14	73, 51.05
<i>Education</i>			
Primary	36, 25.90	33, 23.57	35, 24.48
Junior high	52, 37.41	57, 40.71	56, 39.16
Senior high	30, 21.58	28, 20.00	29, 20.28
University and above	21, 15.11	22, 15.71	23, 16.08
<i>Health insurance</i>			
Urban employee	42, 30.22	42, 30.00	45, 31.47
Residents	92, 66.18	94, 67.14	98, 68.53
Commercial	4, 2.88	1, 0.71	0, 0
Others	0, 0	2, 1.43	0, 0
No	1, 0.72	1, 0.71	0, 0
<i>Ethnicity</i>			
Han	82, 58.99	82, 58.57	97, 67.83
Minor	57, 41.01	58, 41.43	46, 32.17
<i>Disease experience of self</i>			
Yes	15, 10.79	20, 14.29	17, 11.89
No	124, 89.21	120, 85.71	126, 88.11
<i>Disease experience of a family member</i>			
Yes	53, 38.13	48, 34.29	51, 35.66
No	86, 61.87	92, 65.71	91, 63.64
<i>Caring experience</i>			
Yes	51, 36.69	61, 43.57	50, 34.97
No	88, 63.31	79, 56.43	93, 65.03

*Hukou composition was different among the three methods at the 0.05 level using Chi² test

of inconsistencies but a similar frequency of severe inconsistencies compared to the cTTO method.

The two non-iterative TTO methods (nTTO and oTTO) produced overall mean values that were similar to the cTTO method, with nTTO producing the lowest mean value (0.222) and oTTO producing the highest (0.266). However, these mean values were not statistically different from each other.

The mean values by health states across the three methods are shown in Appendix 3. On average, all three methods produced similar mean values for each health state, with oTTO producing higher values across the severity spectrum. Additionally, both non-iterative TTO methods produced a similar number of negative values compared to the cTTO method. Figure 1 displays the value distribution of these three methods at the aggregate level. Over the valuation space, there are 41 distinct values (ranging from -1 to 1, with increments of 0.05), and all possible values have been selected using the three methods. Based on predefined criteria, there were 3 spikes for the cTTO method, 1 spike for the nTTO method, and 6 spikes for the oTTO method. The number of gaps was 6, 7, and 11 for cTTO, nTTO, and oTTO, respectively. More integer values were observed in the oTTO arm and a higher percentage of -1 was observed in the nTTO arm (Fig. 1). All methods displayed a clear pattern of two modals on both the positive and negative segments of the scale. The number of unique values provided by each respondent using the cTTO method was higher compared to the number of unique values provided by the respondents using the nTTO and oTTO methods. However, the value ranges of the three methods were similar.

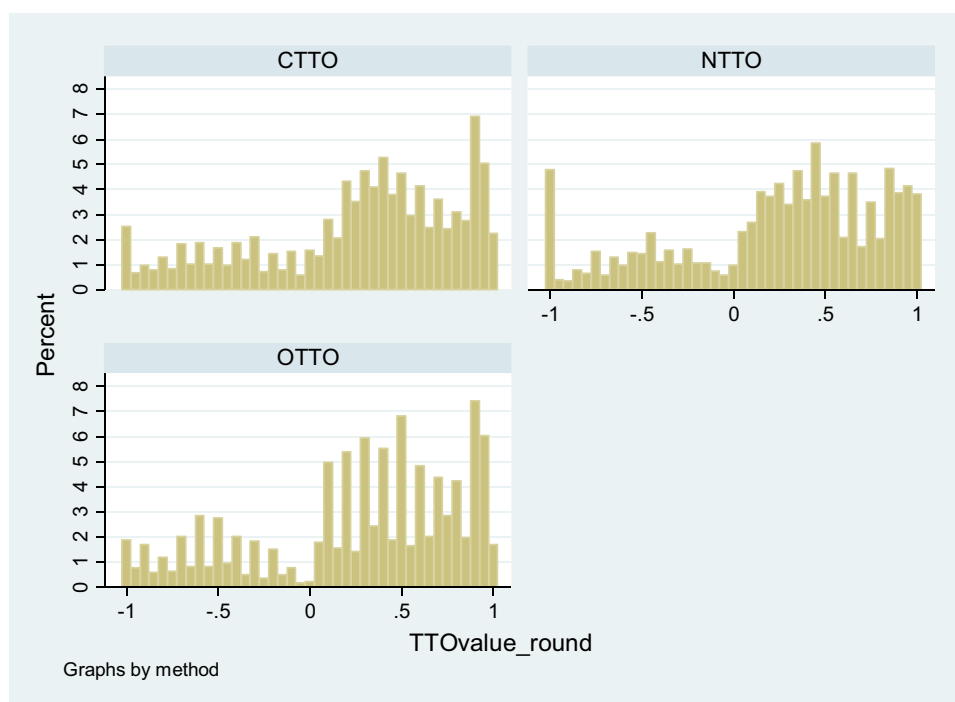
Table 4 displays the coefficients and performance of the CALE model for the three TTO methods. The results indicate that the cTTO method outperformed the other two TTO variants in all four criteria, and the nTTO method performed better than the oTTO method in all four criteria.

Discussion

In this study, we compared two alternative TTO methods with the traditional cTTO method. Unlike the cTTO method, which uses an iterative process, both oTTO and nTTO use a fixed procedure. Both non-iterative TTO methods showed potential for valuing EQ-5D health states, although their data distributions and modelling performance were inferior to the cTTO method. However, both methods demonstrated the advantage of requiring less time to complete the practice and formal EQ-5D-5L health-state valuations, which reduces the overall administration burden. It should be noted that the less time needed for administering these two TTO methods maybe explained by the relative lenient QC process, i.e., we did not specify the minimal time needed for explaining the wheelchair example and completing a single TTO task. The mean values, percentage of negative values, and value range obtained through all three methods were not statistically different. The overall performance of these three TTO methods can be summarized as below. Note this oversimplified overview needs to be interpreted carefully given that some differences were not significant, and the judgement of value distribution was more subjective.

Table 3 Administration burden, logical inconsistency, and value distribution

TTO performance measure	nTTO <i>n</i> = 139	oTTO <i>n</i> = 140	cTTO <i>n</i> = 143
	Mean, SD	Mean, SD	Mean, SD
1. Administration burden			
Practice time (minutes)	10.22, 3.64	9.48, 3.15	12.58, 4.57
<i>P</i> value	0.000	0.000	
Real task time (minutes)	23.01, 9.50	23.76, 9.53	29.38, 9.16
<i>P</i> value	0.000	0.000	
Overall interview time (minutes)	33.89, 10.23	33.28, 10.65	43.68, 11.99
<i>P</i> value	0.000	0.000	
2. Logical inconsistency			
Respondents made severe inconsistency (> 0.5)	6	7	2
<i>P</i> value	0.143	0.084	
Respondents made inconsistency (> 0)	39	51	20
<i>P</i> value	0.004	0.002	
3. Value distribution			
Mean observed value	0.222, 0.582	0.266, 0.577	0.256, 0.567
<i>P</i> value	0.418	0.613	
Percentage of negative value	27.87%	27.55%	27.01%
<i>P</i> value	0.677	0.375	
Mean number of unique values provided by respondent	10.59, 2.06	11.29, 2.32	12.06, 2.35
<i>P</i> value	0.000	0.006	
Mean value range by respondent	1.72, 0.27	1.63, 0.31	1.69, 0.24
<i>P</i> value	0.143	0.119	

Fig. 1 Histograms of TTO values by meth

- 1) administration burden: $nTTO = oTTO > cTTO$;
- 2) logical inconsistency: $cTTO > oTTO = nTTO$;
- 3) value distribution: $cTTO > nTTO > oTTO$;
- 4) modeling performance: $cTTO > nTTO > oTTO$.

It is worth mentioning that while the cTTO method was carried out according to the EQ-VT protocol V2.0 [2], the QC procedures for the non-iterative methods were less stringent. The QC reports were created mainly to identify

Table 4 Modeling results and performance

	cTTO	nTTO	oTTO
Model coefficients: coefficient*, standard error			
Constant	0.928,0.022	0.905,0.024	0.949,0.026
MO#	− 0.349,0.020	− 0.352,0.022	− 0.345,0.024
SC	− 0.236,0.019	− 0.252,0.022	− 0.232,0.023
UA	− 0.326,0.019	− 0.305,0.022	− 0.259,0.023
PD	− 0.468,0.020	− 0.474,0.023	− 0.445,0.024
AD	− 0.300,0.019	− 0.285,0.022	− 0.288,0.024
L2	0.088,0.028	0.136,0.030	0.168,0.034
L3	0.259,0.024	0.252,0.026	0.355,0.029
L4	0.718,0.024	0.638,0.026	0.746,0.030
Model performance			
AIC	1454.01	1762.175	2057.787
BIC	1505.337	1813.23	2108.929
MSE	0.112	0.132	0.151
Adjusted R^2	0.628	0.573	0.514

#MO for mobility, SC for self-care, UA for usual activities, PD for pain/discomfort, AD for anxiety/depression, L2 for level-2, L3 for level-3, L4 for level-4, AIC for Akaike information criterion, BIC for Bayesian information criterion, MSE for mean squared error

*All coefficients are significant at 0.01 level

and address interviewer effects, and only to flag protocol violations in terms of worse-than-dead explanation and occurrence of logical inconsistency, as the appropriate amount of time for the wheelchair examples in these methods was not known. This also suggests that the data collection using these new methods may require less stringent QC procedures.

In this study, both non-iterative TTO methods presented challenges for respondents. The oTTO method posed difficulties for interviewees when asked to specify the exact trade-off years during the final step. To overcome this, interviewers sometimes had to suggest different values with the respondent based on the two initial questions. Moreover, the oTTO method tended to produce integer values as respondents were more likely to use full years (see Fig. 1). This response pattern might have led to imprecise values which caused more logical inconsistency (see Table 3) and poorer modeling performance compared to the cTTO and nTTO methods (see Table 4). On the other hand, the nTTO method experienced a value clustering issue at -1, as observed in a previous study [13]. Although this may reflect the true preferences of the respondents, it is also possible that the non-stopping design of nTTO elicited a passive response style in which the respondents simply repeated the same choice without considering their options. To address this issue, future research may consider simplifying the current nTTO design given that the values of the nTTO were derived when respondents indicate the indifferent points or switch to Life B. All the

choices given before or after these indifferent points or switches are not informative.

There are some limitations need to be addressed. First, we did not randomize these three methods over all interviews, which may be an ideal design to minimize the possible ordering effect. The main reason of not randomizing the three methods is that it can be confusing for interviewers to use three different methods for different respondents, which is prone to errors. Second, the sample size of this study was guided by the recommended sample size for estimating EQ-5D-5L value set [16], that is, 100 observations per health state. This sample size may not have sufficient power to detect small but important differences such as the inconsistency rate across these methods.

In light of the high costs associated with collecting data using the conventional cTTO method, developing alternative TTO methods that can provide data of comparable quality while reducing costs is of great importance, especially for countries with limited resources. The two new TTO methods tested in this study each have their own strengths and weaknesses. For example, the oTTO needs the least time to explain and has the largest value range, while the nTTO produces a smoother distribution and produced more distinct values. Considering their different merits and flaws outlined in the earlier part, future research may create a hybrid TTO method that incorporates the best features of both methods. For instance, we could utilize the initial two questions in the oTTO method to establish an interval and then apply the nTTO procedure within that interval. This approach would reduce the number of questions required, as well as make the questions easier, compared to the oTTO method. For example, in the BTD scenario, if at year 4, a respondent prefers Life A, Life A can be shortened to year 1, and if the respondent prefers Life B, we can determine the indifference point is between 1 and 4 years. Then, the respondent could be asked to perform two TTO tasks in which Life A is 2 years and 3 years, respectively. Such new methods could be developed and evaluated in future studies.

Conclusion

The results of this study showed that the two alternative non-iterative TTO methods posed less administration burden when a loose QC procedure was applied. When compared with the cTTO method, these two methods clearly produced inferior TTO data, but the overall quality of the data was acceptable. There is a clear trade-off between optimal data quality and less administration burden when choosing among these TTO methods. These alternative TTO methods could be useful for valuing EQ-5D health states, particularly for methodological studies that the concerns of administration burden and feasibility may outweigh the concern of data

quality. Further research is needed to more thoroughly test these methods and explore new methods that combine the best features of these methods.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10198-023-01647-x>.

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Data availability The data that support the findings of this study are available from the corresponding author, Luo, N, upon reasonable request.

Declarations

Conflict of interest Dr. Yang reports grants from The EuroQol Research Foundation, during the conduct of the study; grants from The EuroQol Research Foundation, outside the submitted work; and Dr. Yang is an EuroQol member. Dr. Rand reports grants from The EuroQol Research Foundation, during the conduct of the study; grants from The EuroQol Research Foundation, outside the submitted work; and Dr. Rand is an EuroQol member. Dr. Stolk reports grants from The EuroQol Research Foundation, during the conduct of the study; grants from The EuroQol Research Foundation, outside the submitted work; and Dr. Busschbach reports grants from The EuroQol Research Foundation, during the conduct of the study; grants from The EuroQol Research Foundation, outside the submitted work; and Dr. Luo is a member of the EuroQol Group. Dr. Luo reports grants from The EuroQol Research Foundation, during the conduct of the study; grants from The EuroQol Research Foundation, outside the submitted work; and Dr. Luo is a member of the EuroQol Group.

Ethical approval This study received exemption from the Guizhou Medical University Ethic Committee because this human subjects research qualifies as no risk or minimal risk to subjects and all of the research procedures fit within one of the exemption categories in Guizhou Medical University IRB regulations.

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